and put it in place, there won't be a wet spot left of any of us," theatrically declared Moskalenko, known among the troops by his nickname, "General Panic." There was yet another problem with the R-16: the fuel had to be drained from the rocket's tanks if it was not launched, as its unstable components were too corrosive to remain there longer than a couple of days. "According to American experts," said Moskalenko, "the solid-fueled Minutemen can remain constantly ready for years."[9]

The chief designer of the R-16 missile, Mikhail Yangel, who had survived the catastrophic explosion of his rocket in October 1960, was now working hard to improve its design but could not promise quick results. When Yangel went to the podium, he told Khrushchev that the R-16 was the last missile of the generation that began with Sputnik. The Minuteman had changed the game. The Soviet Union needed a new generation of missiles that could stay ready for launch at any minute. Disturbing news was coming not only from Yangel and his design bureau in Dnipropetrovsk, where Khrushchev had boasted that missiles were being mass-produced like sausages. Yangel's competitor Sergei Korolev, the father of the Sputnik, was struggling with his own problems. His rocket, the R-9, also used liquid fuel—a mix of kerosene and oxygen stored at low temperature. Loss of oxygen from the fuel tanks required constant refueling of the missile. If the R-16 had already been tested and was ready for deployment, the R-9 was still undergoing tests, each of them uncovering new problems with Korolev's rocket.[10]

A further problem was the virtual absence of battle-ready long-range missiles. Very few of the missiles that Khrushchev had at his disposal were intercontinental ones capable of reaching the United States. Korolev's R-7A rockets could do so, but there were only four of them, and they were unreliable and badly outdated, requiring twenty hours for fueling, which made them a perfect target for US bombers. Yangel's R-16 was much better but had just entered the deployment phase, and tests of the silo-based models had just begun. Altogether the Soviets had only a few dozen R-16s, vulnerable, as Moskalenko had explained, to enemy missile attack and incapable of being launched as part of a second strike. The Soviets thus had little to deter a possible attack from the United States.[11]